

SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Closed-loop architecture enabling LLM agent skills to continuously evolve by aggregating cross-user interaction trajectories

Agent Skills Remain Static After Deployment

Current Paradigm: Static Skills

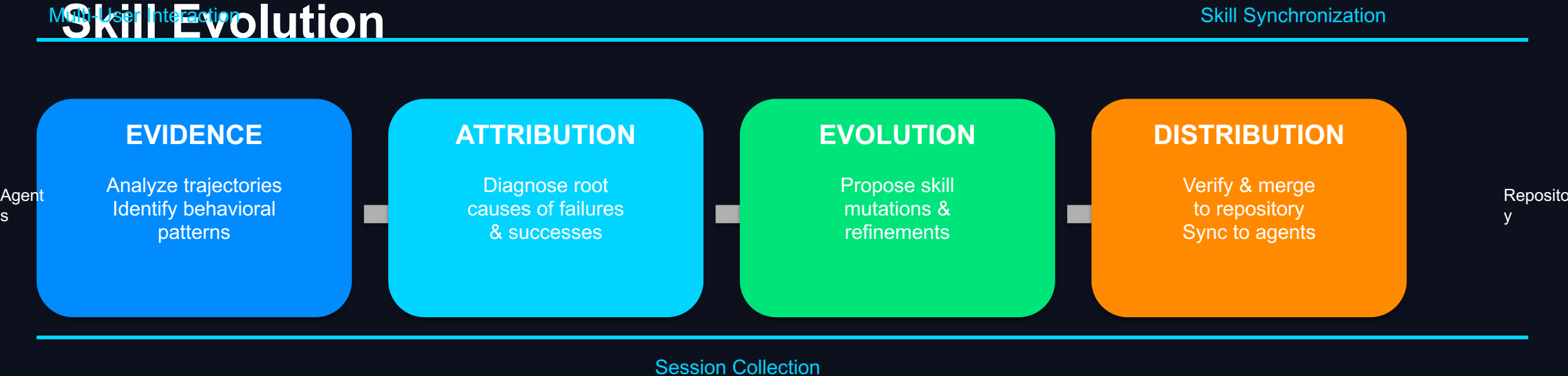
- Skills manually installed from centralized hub
- Solutions from runtime never persist beyond session
- Same failures rediscovered across users repeatedly
- No system-level knowledge accumulation
- Each user forced to solve same problems independently

SkillClaw Paradigm: Evolving Skills

- Skills continuously updated from runtime evidence
- Cross-user trajectory aggregation drives improvement
- Shared evidence base enables collective learning
- System-level knowledge accumulates over time
- Solutions benefit entire agent ecosystem

Key Insight: Individual interactions rarely generate enough signal to separate generalizable improvements from idiosyncratic fixes

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



Agentic Evolver performs open-ended reasoning over evidence — not predefined update rules

Compatible Systems: OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw

Three-Stage Pipeline: Evidence → Attribution → Evolution

1

EVIDENCE

Analyze trajectories to identify behavioral patterns

- Aggregate session trajectories preserving action-feedback chains
- Group by skill under diverse contexts
- Identify success patterns and failure modes
- Cross-user aggregation reveals complementary signals

2

ATTRIBUTION

Diagnose root causes of failures and successes

- Determine why a skill succeeds in one context but fails in another
- Separate generalizable improvements from idiosyncratic fixes
- Identify skill boundary conditions
- Single-user signal is insufficient for reliable attribution

3

EVOLUTION

Propose and refine skill mutations

- Open-ended agent reasoning — not predefined rules
- Propose skill refinements or new skill creation
- Conservative verification prevents regressions
- Updated skills merge to shared repository

Key: The evolver is an autonomous agent performing open-ended reasoning — not a rule-based system

Cross-User Aggregation Exposes Skill Behavioral Boundaries

Compatible Claw-Style Systems

- OpenClaw
- CoPaw
- IronClaw
- PicoClaw
- ZeroClaw
- NanoClaw
- NemoClaw

Why Cross-User Aggregation?

Different users exercise the same skill under diverse contexts

Complementary views reveal both success conditions and failure modes

A single user rarely generates enough signal to separate:

- Generalizable improvements (benefit all users)
- Idiosyncratic fixes (session-specific, not transferable)

Grouped trajectory evidence enables reliable skill boundary detection

Without aggregation: failures re-occur, solutions not accumulated

Result: Collective evolution enables shared skill ecosystem improvement that no single user could achieve alone

Multimodal Creative Task Pipeline — 6 Days of Evolution Attempts

1 of 6

Accepted

Day	Candidate Skill	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Reject	Keep current best pool
5	multimodal-creative-task-pipeline (impr.) + validate-required-input-files	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Reject	Not admitted to next cycle

Conservative verification process: Only 1 of 6 candidates accepted over 6 days, preventing regressions

Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

3 of 6

Accepted

Day	Candidate Skill	Validator	Next-Day Action
1	git-push-with-auth-fallback	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Accept	Keep current best pool
4	(none)	Accept	Continue same best pool
5	git-push-with-auth-fallback	Reject	Keep current best pool
6	git-push-with-auth-fallback	Reject	Not admitted to next cycle

Iterative refinement: 3 accepted updates before plateau — successful skill improvement through evidence-driven evolution

Three Properties That Distinguish SkillClaw

#1

**COLLECTIVE
EVOLUTION**

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem

#2

**FULLY
AUTOMATIC**

Skill evolution driven by runtime interaction without manual curation or explicit user intervention

#3

**AGENTIC
EVOLUTION**

Skill updates produced through open-ended reasoning rather than predefined update rules

Evaluation: WildClawBench with Qwen3-Max backbone shows substantial task improvements

Implications: Building continuously improving agent systems that learn from collective multi-user experience